

Cognitive Grammar analysis of Compromise Forms in Bilingual Speech at an Idiolectal level

Abstract

This paper presents evidence against matrix-language approaches to code-switching by applying Ronald Langacker's Cognitive Grammar framework to bilingual pedagogical discourse. Analyzing seven classroom lectures from a multilingual Indian instructor, the study examines non-standard compromise forms, focusing integration of construction schemas through extension relations at the level of the individual structured inventory. Rather than demonstrating a shift between bounded linguistic systems as separate mental spaces, the data reveals a fluid, integrated structured inventory where constructions are selected based on cognitive economy and conceptual negotiation. The results show that non-standard syntax emerges from the convergence of entrenched structures across languages, serving to reduce processing cost during spontaneous delivery. Ultimately, this study supports a usage-based, construction-driven model of bilingual competence.

Keywords: code switching, bilingualism, cognitive grammar, compromise forms, integration

1.Introduction

Code Switching has been investigated through various approaches and models. Since code switching research is done on a data basis, many types of specific instances of code switching phenomenon have been identified. Some assume a dominant matrix structure and show insertions as a foreign element. These approaches define language with the help of the definition of a standard unit of language established on a broad socio- political or geo-cultural levels.

A significant number of usage based analyses of the phenomenon have also been done to examine the relation between code- switching and cognition and the correlation with Executive functions. Bilingual construction grammar, and cognitive grammar defines a unit of language as a network of constructions which vary along the ranges of complexity and schematicity. Thus, it does not use the affiliation to a standard unit as the basis of analysis or a strictly bounded mental space within an individual cognitive system. However, most of these experiments are done within the domain of psychogrammar while dominant

traditions hold a strict dichotomy between performance and competence. Cognitive grammar emphasizes that this is a false dichotomy.

However, standardised and codified forms are used in the two cases for standard to target comparison between the structures to analyse the change in the function of the predicate within the clause at the idiolectal level where diachronic and synchronic pressures are naturally shaping linguistic structures along with the functional requirement of conceptual negotiation within the limited time of a classroom lecture. At the specific level, the neutral average access time indicates a single structured inventory used in conceptual negotiation where compromise constructional schema with elaborated semantic poles show which serve a specific function in the clause.

2. Literature review

Categoriality has presented many questions since antiquity. Post the development of the prototype model, and the widening of data and corpus about the languages of the world, categoriality has been examined quite exhaustively by Hopper and Thompson (1984). They consider discourse function as a primary determinant of categoriality, showing how the structure of the construction schema depends on communicative intent and discourse function.

Hopper and Thompson (1984) further discuss the loss of categoriality for N and V as a part of discourse function. The loss of modifier and grounding elements has been observed by them as categoriality loss, including what is identified as schematization in cognitive linguistics. This is not in contrast to cognitive linguistic understanding of elaboration, extension and correspondence relations. Loss of categoriality has been explained through the prototype model : losing the markers that make a form or structure a good representative of a category in order to become a marker of another category. An exhaustive cross linguistic examination shows, nuanced contrast between referentiality, manipulability, anaphoric mention, loss of sortal markers and sensitivity to essives; including verbal incorporation of noun like elements even causing mutilation and unverbation. They also show that verbs become sensitive to markers for existence, purpose by losing categoriality using predicative autonomy as the prototype for the verb to create further nuance structured by discourse functions.

Hopper et al. goes on to argue that the more a category introduces a manipulable entity into the discourse, the more marked and categorial it has to be. This eliminates information

which refer to a previous mention instead of directly referring to the entity such as an anaphoric mention: the most extreme form of decategorialization being ellipsis. Borderline entities like adjectives and other modifiers are decategorialized N or V and serve specific discourse functions. For the verb, it is predicative autonomy. The optionality of marking serves a discourse function.

2.1 Hindi to English Code switching and mixing from the structural perspective

Khubchandani (1985) in a study on diglossia observes that in the South Asian context, especially with respect to Hindi and English, which acquire specific functions for intra-group communication, there is a “fluidity of codes”.

including the phenomenon of bilingualism are instruments of linguistic stratification for the speech community. Apart from language attitudes; settings, channels of communication, situation and communication tasks modulate speech behaviour. While he leaves open the question of speech behaviour reflecting adequacy versus speech behaviour reflecting variation which is acceptable considering the lack of a standard even within a specific geocultural or geopolitical context; he does note that that in narrow khariboli regions, speakers of urban and semi urban centres in the fluid zone do not acquire control over the language of the region because of pressures of urbanity or modernity created by a prestige language which is associated with formal literacy. While discussing the case of Hindi Urdu Punjabi, where HUP is the prestige variety in competition with non khariboli languages; he shows that the switching of linguistic codes from the native speech to Hindi Urdu becomes similar to the switching of styles in a monolingual situation. He also categorizes anglicized Hindi as a middle brow Hindi and attributes it to increasing impact of technologization and urbanisation of Indian society.

Kumar (1986) in a study of the formal and functional properties of English to Hindi code switching; notices that the phenomenon of code switching is based on discourse function and influenced by what structural linguists understand as extra linguistic factors within the domain; such as topic or interlocutors or setting etc. The definition of code switching provided by him is-

Code- switching, “which is influenced by extralinguistic factors such as topic, interlocutors, setting etc, is the alternate use of lexical items, phrases, clauses, and sentences from the non native language(English) into the system of the native language(Hindi) (Code mixing is here subsumed under code switching) ” (Kumar, 1986, p. 196)

Since it is the discourse functions which lead to code switching, it is thus assumed that repeated instances of code switching that lead to the artifactual codes becoming mixed in a certain pattern at such a scale that it becomes conventionalised: the units of language being decided at the

socio-political and geo-cultural level. The study shows code switching patterns that are relatively regular enough to be available in film magazines which are brought out in Hindi and have a wide circulation in all parts of India. He noticed that certain categories do not switch and among them are determiners even though head nouns and modifiers may be switched. The reason given is the determiners are closed class categories and situated more towards the grammatical end of the lexical-grammatical continuum. This is also true for demonstratives, pronouns, conjunction, articles, interjections and bound morphemes. In compound nouns, both constituents are predicates from the same artifactual unit. Genitives, pronoun subjects and pronoun objects don't show switching.

It is interesting to note that construction schemas like the position of the reference point follows the standard predicate. The example given is-

‘Meeting of hero-heroine’ and not ‘*meeting ka hero- heroine’
‘hero heroine ka Milan’ and not ‘*hero heroine of Milan’

Thus, the predicate and the construction schema have such a correspondence that when the construction schema of 1 standard artifactual unit is accessed, the predicate of the unit is automatically activated and accessed.

This is a bilingual pattern that has been noted.

There are certain operators in Hindi, which are both conceptually loaded and have grammatical function. These are dependent, profile determinants at a higher compositional level.

*Create do

*Nirmān do

The switching of the tense operators. He notes that Hindi tense operators cannot be separated. The temporal grounding and the aspectual state cannot be separated.

This is further strengthened by an argument for unique constructions schemas like reflexives which do not shift like tense aspect markers although Kachru does point out the adoption of reduplication of an English predicate within a code mixed variety (Kachru, 1976)

English items, adopt the morphology of the Hindi like the forms: faailō, kampaniyāā

Lastly, the English predicates are shifted to the end of the clause.

“Abhi mujhe mili, yah shocking news”

I got this shocking news just now

" Film ka publicity manager Jo tha, voh jyada Safal nahin hua”

The publicity manager of the film did not succeed much

Although he recognizes the role of the participants, setting and level of proficiency in these generalizations.

Kachru (1976) within the backdrop of language dependency within multilingual society posits an inner language and an outer language structure with respect to the South Asian periphery. In the elaboration of functional roles or roles in a multilingual person's "restricted or extended spheres of linguistic interaction": they find patterns of two types. Within the domain of the reference point which is South Asia heterogeneity is being neutralized through linguistic convergence while the second type involves imposition of languages from outside the South Asian periphery.

Although, most of the terminological signposts used by Kachru, like Aryanization and dravidization and Persianization maybe arguably assessed as reductive with respect to linguistic phenomenon from the diffusion perspective; the formal manifestations narrowed as a character of standard code mixing taken to account role dependent and function dependent linguistic phenomenon. Language use and the form of language both belong to a framework made by the concept and the context of the situation forming contextual artifactual units which become very important in the era of digital media and digital news.

Kachru, regards code mixing as an extension of borrowing. Code mixing phenomenon identification involves necessarily a Disystem such that another code can be developed comprising the formal features of the two codes where he notes that a particular type of lexical and grammatical cohesion is associated with a specific type of discourse or register as a speech function. Unit insertion of both NP and VP, mixing within units to create hybridization, insertion of an entire sentence structure, insertion of idioms and collocations, insertion of inflection, attachment and reduplication are discussed including the fact that predicates of a standard are readily accessed when the construction schema is accessed through secondary activation as shown by the patterns of constraint. From the construction schema of the noun phrase identified as the rank shift constraint, position before or after genitive, and tendency to delete agent in passive constructions, Kachru argues that predicates and constructional schemas of a single standard unit: has the activation of one resulting in the activation of the other.

Gardner-Chloros and Edwards (2004) points out the variety in code switching phenomenon which compels a reconceptualisation of the related senses of grammar as competence or an external, self contained artefactual system. While they introduce four senses through which the concept of grammar is understood; they question the assumptions behind any undertaking to examine the code switching phenomenon. They recognise that the role of grammar is mediated by poly-idiolectal repertoires of the speaker categorizing the usage event with respect to a specific context thus problematizing the quest to find a universal rule that accounts for the

variability of the phenomenon. Variability is found at several levels from inter community to even within the speech of the single individual within a single conversation.

The structure of the switching may or may not find a native model in any of the artifactual units that are being considered to be mixed through switching in this kind of verbal behaviour. They also compare approaches which blur the line between code switching and borrowing in the construction of compromise forms like Treffers-Daller and approaches like Poplack which draw a clear distinction between codes switching and nonce borrowing.

Structural approaches whether constraint based or matrix language based presuppose underlying structures when working on congruity as predictive of points of switching. This is also similar to the alternation proposed by Muysken in 2000 when he speaks about compatibility of two units of artefactual language or a point of equivalence where switching occurs whether insertion or congruent lexicalisation. In 1998, Sebba argued the objective measure of equivalence as constructed by individual speakers. Matrix language approaches consider a unit of structured inventory to be more activated in the brain from the perspective of psycho grammar while from the social perspective; the matrix language is an unmarked choice which again depends on sociolinguistic circumstances. On the whole, assessments of code mixing or code switching despite the variation in the scope of the definition; present a problem for analysis at the level of language as external to the individual because of the social, personal and interactional reasons which are the primary determinant of the phenomenon of switching (p.123).

3. Code switching and cognitive grammar

Cedden et al. (2024) map out the complexity of the approaches to the code switching phenomenon and the historical evolution of research on code switching. Muysken's code switching categories: alternation, insertion and congruent lexicalisation, play a significant role in these approaches. In 2017, Jylkkä et al found, through a self assessment questionnaire in Finnish-English code switching, faster times in the mixed language block which gives an advantage of mixing. Something similar was found by Hofweber, J., Marinis, T. and Treffers-Daller, J.

In 2019, Gross, Lopez, Buac & Kaushanskaya (2019) on Spanish-English code switching found cognitive control and language ability modulate processing cost or constructional effort for the coding of the target structure. In fact, Jylkkä, J., Laine, M., & Lehtonen, M. (2021) found that switching improved executive functions. A correlation between attentional states has been found with code switching by Green and Wei in 2014, Green, 2018) where alternation involves narrow attentional state compared to integration. However Hofweber (2020b) shows that dense code switching will have the grammatical and lexical features of both languages activated at the same time. Many studies involve a dominant language or a matrix language which is a prototypical representation of an artefactual unit.

The Matrix language model is questioned by Wasserscheidt (2020). He shows that the assumption of any theoretical unit above the level of construction is gratuitous in the analysis of code switching. It was introduced in 1982 by Joshi and 1985 by Klavans into modern bilingualism research. Muysken in 2000, had emphasized that the phenomenon is always asymmetric where one unit of language is thought to determine an underlying autonomous syntactic structure which is activated without the secondary activation of the units that instantiate them from within the structured inventory of the artefactual unit. This is supported by Backus(2010) and Chan(2010: 187). Based on the assumption of native code and foreign code, Johanson(1998:327) states that copies of elements are inserted into some basic frame. This is analysed through the control of the head outside the immediate structure. If a frame is assumed, early system morphemes follow and thus the assumption of the matrix language.

The construction grammar view, upholds the concept of a structured inventory which is made of constructions which exist at varying degrees of schematicity. They are related to each other by forming a network of conventionalised and complex combinations of form with meaning along with usage conditions. These have to be entrenched through learning and relearning. In the cognitive grammar model, units are learned through decontextualization which aids generalisation. These constructions are in a continuum with respect to degree of complexity and schematicity. Thus, the unit of language becomes a mental unit, a construct defined as

“Part of a heterogeneous network of formally overlapping and frequently co-activated units with similar contextual or pragmatic information”.

The cognitive grammar framework proposed by Ronald Langacker does not differentiate meaning from conditions of use for grammatical description considering the fact that the target unit is coded to categorize a specific usage event, specified upto the situation of utterance.

Bilingual construction grammar proposes that switching should be possible at the same places where the structure of a unit should combine. Thus, the constraints within the unit should not be violated as found by Muysken (2013:715). What Wasserscheidt notes is that, there is an assumption of a genuine symbol of a conventionalised unit. It is here that Backus (2010) becomes relevant who notes that both synchrony and the diachrony are relevant for codes switching analysis.

Insertional switch is in a continuum with lexical borrowing with respect to the synchrony diachrony dichotomy. Calques or semantic extension or narrowing is in a continuum with lexical change. Interference is in a continuum with structural change or borrowing. They study the switch between monolingual Turkish and immigrant Turkish to assess the structure of the change.

Wasserscheidt notes that Kecskes in 2006 had established that code switchers, especially in minority languages, may not possess separated linguistic systems, but they possess blended concepts and constructions. Compound verbs, portmanteau constructions, double marking of grammatical functions. He notes McSwan(2005) and Auer(2007);who show that languages cannot be the basis of linguistic analysis and language should not be the starting point of how we account for bilingual data where the term bilingual refers to multiple units that are external to the organism. Each community of praxis above the individual, has an own sub network within the standardised network of artefactual units; where the standardised network itself forms a continua at a larger scale. Most importantly, Kecskes is noted to point out that in the case of bilinguals and multilinguals; they do not select languages but rather constructions from a structured inventory. Languages as units are not competing for attention, but rather words and constructions are competing for attention. This can account for alternation, and various kinds of code copying with respect to artefactual units while also explaining compromise forms and integration at varying levels. In addition to Meyers Scotton's (2000) observation that a grammatical frame prevails, Wasserscheidt adds that this is true but a frame may be unaffiliated with any particular language unit. Whether there is any coordination of separate mental spaces at the level of unit of structured inventory has not been established yet. (Fauconnier (1985/1994)

Code mixing, code switching, insertion or alternation, loan translations, code copying or borrowing form continua along both the lexical and grammatical dimension and along the synchronic and diachronic change of language.

However, a significant gap remains in exploring the modification of construction schemas without the assumption of another unit as a limiting case or assuming constraints. This study attempts to examine the construction without assuming a limiting case. While the comparison of structures may yield a value of similarity, no identity inclusion relation at the semantic pole is assumed.

4 Methodology

This study uses the cognitive grammar framework proposed by a Ronald Langacker (Langacker,1991, 2001, 2002, 2009, 2013, 2017, 2020) to describe certain compromise forms in bilingual speech using predicates and constructional schemas of varieties of standard Hindi and varieties of Indian English. This paper tries to show how extension relations are formed based on the functions of the particular predicate in a clause between elements of two artefactual structured inventories within the structured inventory of a single speaker within a particular context which then further decontextualizes during the entrenchment process and how these integrated compromise forms are used in conceptual negotiation within the classroom situation.

7 class room lectures(approximately 35-55 minutes each) provided in a non standard lect given to first year engineering students in an extra departmental psychology course are recorded and analysed. This paper attempts to discuss some of the notable compromise forms to show the change in categoriality with respect to the artefactual unit. To this effect, no Matrix language or unit affiliation of the frames has been assumed for the analysis which focuses on the constructions. The constructions form a continua with standard structural inventories therefore a standard to target comparison for the structure yields an approximate value of the altered categorial state.

4.1 Subject summary and other details

The subject is an assistant professor of psychology in a reputed National engineering college in India which is a multilingual country. He is born in Varanasi. He has completed education from Delhi. The Hindi spoken in Varanasi is altogether a different variety, including sociolects and registers than the Hindi spoken in Delhi. He is working in West Bengal although within an isolated structural unit designed to house students and teachers from all over India within a residential campus.

4.2 Analysis

The language used in this specific idiolect as a type of pedagogical lect contains both the predicates and constructional schemas from the artifactual Hindi varieties and the Indian English varieties.

With respect to any artifactual variety, the coded target structures are partially schematic at the higher compositional level, fully sanctioned by the artefactual units at the lower compositional level.

The target structures achieved by the process of coding provide an insight into the patterns of mixing

Complex categories are formed when a member of a category, where instances belong within a continua of the prototypical to the deviant: is a member of multiple type spaces simultaneously existing within any point in the range that forms the type space of the category.

4.2.1 Limiting scope of universal quantifier to maximal extension within bounded type space

“So this all has been done” (3.00)

To understand how the proximal deictic functions as a superordinate grounding element in the inverted 'this all' construction, it is first necessary to examine how its baseline grounding scope is partitioned and supplemented by the qualitative modifier 'particular' within the speaker's idiolect.

The proximal deictic has been used to refer to abstract substances which are bounded configurations.

For example:

“This particular task”

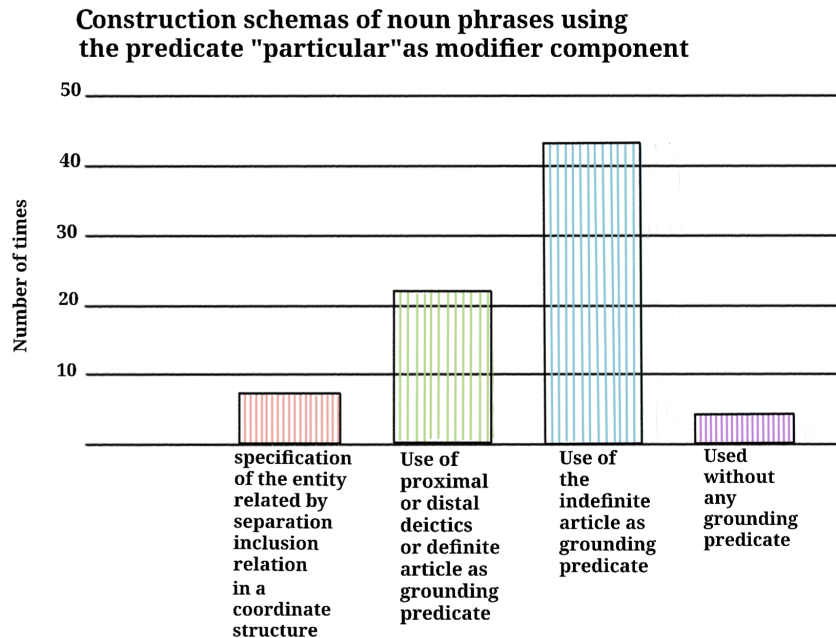
The type task at the semantic pole, is a configuration of participants and events situated in conceived time. The proximity to current reality is due to sequential scanning through discourse space. It is the proximity to structures in adjacent attentional frames which results in the assumption of mental contact.

The adjective predicate "particular" is conceptually loaded and domain specific as per categorization of usage events. It is, as defined in the English dictionaries which are commonly referred to, in the Indian curriculum-

Oxford Advanced Learner's Dictionary (OALD)	Used to emphasize that you are referring to one specific person, thing, or advantage and not others; (of a person) very careful about choosing something and not easily satisfied. (Oxford University Press, n.d.)
Cambridge Advanced Learner's Dictionary (CALD)	Special, or this and not any other; used to describe someone who is very careful and chooses only those things that are of a high standard. (Cambridge University Press, n.d.)
Merriam-Webster	Of, relating to, or being a single person or thing as distinct from some or all others; deserving special attention or notice; picky or fastidious. (Merriam-Webster, n.d.)
Collins COBUILD	You use particular to emphasize that you are talking about one specific thing or person rather than others; if you say that someone is

	particular about something, you mean that they have very high standards and are difficult to please. (HarperCollins, n.d.)
Longman Dictionary of Contemporary English (LDOCE)	Used to emphasize that you are talking about one specific person or thing and not others; very careful about choosing exactly what you like and not easily satisfied.(Pearson English, n.d.)

Since the predicate in question within the data, is attributed to an abstract substance which is 'task'; three out of five definitions focus on specificity and uniqueness which is indicated through an exclusive type space that categorically excludes the other which is construed at lower schematic levels for this purpose. This is functionally achieved by "this and not any other" (Cambridge University Press, n.d.). This predicate is a component of a co-ordinate structure that uses a separation inclusion relation with the predicate 'special'. This predicate emphasizes uniqueness. Merriam Webster construes further the predicate 'single' which includes a cardinal quantifier within the conception; thus a bounded mass or count region in some domain. Merriam-Webster further specifically construes the relation between the region and 'some' or 'all others' to be association inclusion relation or separation inclusion relation or identity inclusion relation. 'Of' typically profiles a target within the dominion of a reference point, construing the separation of the target from all the instances within the type space of the maximal extension of the dominion of the reference point which indicates uniqueness. 'relating to' specifies any specific relation holding between the dominion and the target while 'being' construes the identity of the target as a specific part of the dominion of the reference point.



Within the idiolect data, the specification of entities related by the separation inclusion relation within a coordinate structure have instances which are in close proximity and have a topic in the conceptual substrate common between them. An adversative predicate ‘but’ is used in 5 out of 7 instances to modify the scope of (ONE PARTICULAR NOUN) or (A PARTICULAR NOUN) in the semantic pole e.g
 “learning to respond to a particular stimulus but not to stimulus that are similar”

In 1 out of 7 instances, the conjunct predicate is used to add to the scope, the construal of some entities whose participation in the situated event is negated at the higher compositional level
 "To discriminate one particular tone and not other tones”

The standard English constructional schemas are used maximum number of times.
 [[A PARTICULAR NOUN]/ [a particular]] is used the most. The indefinite article profiles a single region with the maximal extension as conceptual base. The predicate ‘particular’ highlights the idiosyncrasy of this region within the domain and with respect to a specific conceptual domain.

With the deictic, it indicates a uniquely bounded region in the CDS. These are also sanctioned by standard English constructional schemas-
 [[THE PARTICULAR NOUN]/[the particular...]] , [[THIS PARTICULAR NOUN]/[this particular...]] , [[THAT PARTICULAR NOUN]/[that particular....]]

The predicate is also used without any grounding element [[PARTICULAR NOUN]/[particular...]] but it is important to note that the noun it modifies is not a replicate mass where quantification and uniqueness is both specified by the predicate.
“To weaken particular response”

Thus, the proximal deictic is not sufficient to specify singularity, speciality and exclusion of the other instances in a specific type space by itself in the idiolect. The predicate ‘particular’ is used to indicate the exclusivity and idiosyncrasy of the instance.

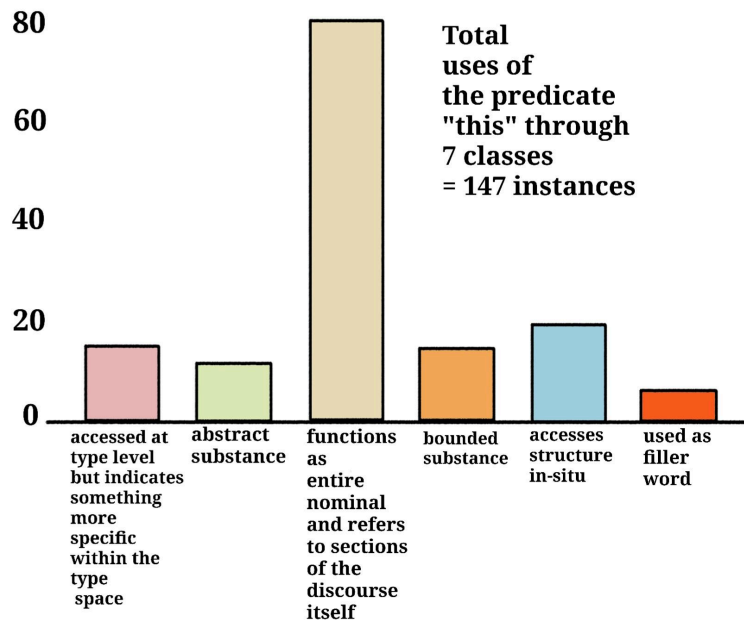
The contrast between “this thing” and “this particular thing” in this idiolect is the additional substructure of exclusivity and idiosyncrasy. e.g “so Watson induced fear in that particular five six year old infant.” This eliminates the semantic substructure of idiosyncrasy from being a structure of the proximal deictic predicate and shows that the idiolect differentiates the semantic substructure of idiosyncrasy from uniqueness.

There is only one instance in 7 classes, where a pronoun serves as a reference point ex- “gives our particular perception”. The predicate ‘particular’ is used in two instances which are phonologically similar in a composite noun phrase without any grounding. In this case, it is idiosyncrasy without invoking uniqueness or singularity of instance.

Ex- “to weaken particular response”

In this case, idiosyncrasy is invoked to isolate a specific instance within a type which functions as a discursive generic or role which serves to isolate an instance at a higher schematic level without a specific grounding element such as the definite article or the indefinite article such that R_T is not required in the conceptual base to profile the uniqueness of the instance. The predicate “particular” operates at the qualitative type level relative to specific conceptual domain whereas the proximal deictic operates at the situational instance level. “This particular task” therefore achieves both situational grounding and qualitative isolation.

When the specific conceptual domain in question is the CDS through which mental contact is achieved, the basis for idiosyncrasy is most often integrated into the sequential scanning of a region within processing time comprising the discourse, which is bounded extra linguistically through the concept of class. However, idiosyncrasy is more lexical than grammatical within the continuum and thus relative to the structure of the conceptual domain within the semantic pole. It should not be confused with grounding, where mental contact is assumed regardless of the conceptual domain in question. However, since substructures inside this conceptual domain have syntagmatic relations with the grounding predicate where elaboration occurs between concepts that are very close in the continua of the type of association inclusion relation, it is important to consider the reason for the need of the predicate “particular” which was not satisfied by the proximal deictic.



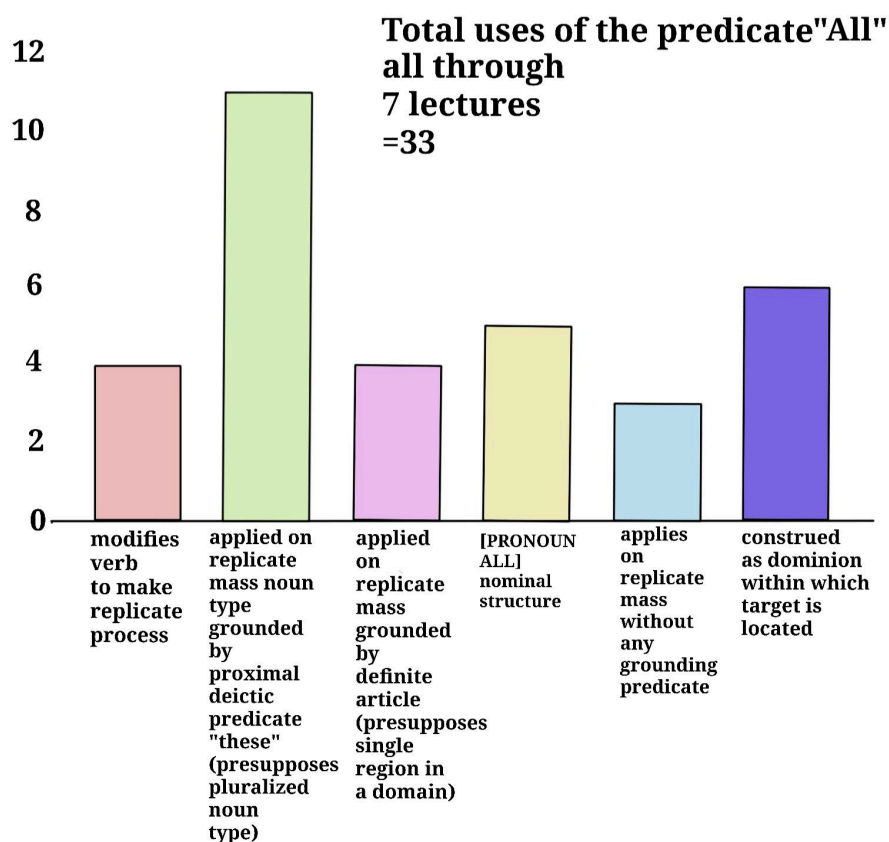
The proximal deictic grounds a nominal and functions to indicate that despite being accessed at the type level, the composite nominal refers to a specific instance within the accessed type space e.g “this experiment” “this much” “this thing”.

It grounds a region in a domain where it can be a bounded region or it can be an abstract substance construed through a nominalization, e.g “this wincing”, “this behaviour”.

It is used as a complete nominal to refer to a section of the discourse where the CDS is exhausted by the body of the discourse and involves scanning through the discourse. This function is utilised the maximum number of times where the noun type remains schematic and mental contact is assumed. The reference point is situated in prior attentional frames where the predicate serves an anaphoric function.

It is also used to access something in the situational context such as " this (NOISE)", " the writing would be something like this(WRITING ON BLACKBOARD).The proximal deictic accompanied by the instructor physically making a noise, or pointing to a writing in the situational context distinguishes its function of referring to adjacent attentional frames to the immediate attentional frame.

This predicate is also used as a filler word to indicate difficulty of access like "this Skinner" "now this" etc. However the type is not idiosyncratic with respect to the domain at the semantic pole. It is used for assuming mental contact, the contrast between the predicates "this" and "these" shows that the former includes the substructure of singularity at the semantic pole along with referring to something situated in-situ.



The idiolect contrasts between "All this information" and "This all has been done".

This predicate is used after the verb, to superimpose the construal of a replicate process when it functions as an adverb. It is mostly used in a nominal structure.

In the Standard English constructional schema, [ALL THIS/ all this], has the same function. The head modifies the type and the determiner indicates which instance of the type is being referred to. The proximal deictic is prototypically often used as an entire nominal where the noun type is unspecified. In that case, the noun remains a schematic substructure within the proximal deictic, which is then grounded by the universal quantifier which indicates that the schematic noun is a replicate mass. The proximal deictic grounds the type, and the universal quantifier profiles the totality of a multiplex entity.

In the Standard Hindi constructional schema [PROXIMAL DEICTIC UNIVERSAL QUANTIFIER/ *yaḥ sab*] or [PROXIMAL DEICTIC UNIVERSAL QUANTIFIER/ *yē sab*], the proximal deictic defines and limits the scope of the universal quantifier by presupposing it. This is a prototypical construction. The head makes specific the type and the modifier indicates which instance of the type is being referred to. A multiplex entity is construed in totality as a reference point before internal heterogeneity as target within the dominion of the entity is scanned to profile the maximal extension including internal heterogeneity within the bounded region.

However, Standard Hindi treats the universal quantifier as having the same function as the absolute quantifier which prototypically permits

[[PROXIMAL DEICTIC][ABSOLUTE QUANTIFIER][NOUN]/ [*yaḥ*] [*tīn*] [*kitabē*]]

but not

[[PROXIMAL DEICTIC] [UNIVERSAL QUANTIFIER] [ABSOLUTE QUANTIFIER] [NOUN]/ [*yaḥ*][*sab*] [*tīn*] [*kitabē*]]

or

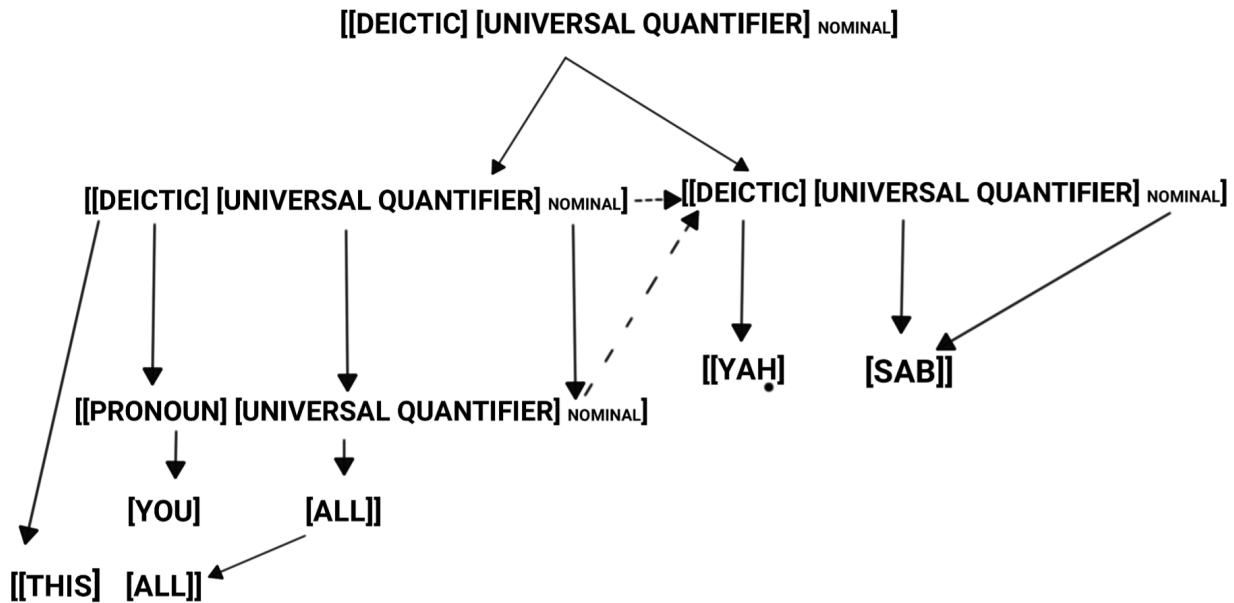
[[UNIVERSAL QUANTIFIER] [PROXIMAL DEICTIC] [ABSOLUTE QUANTIFIER] [NOUN]/ [*sab*] [*yaḥ*] [*tīn*] [*kitabē*]]. The proximal deictic prototypically comes before the universal quantifier.

Standard English constructional schema prototypically permits [UNIVERSAL QUANTIFIER PROXIMAL DEICTIC NOUN/ All these things] or [UNIVERSAL QUANTIFIER PROXIMAL DEICTIC ABSOLUTE QUANTIFIER NOUN/ all these three things]. It differentiates the function of the universal and the absolute quantifier.

The universal quantifier in standard English profiles a mass which coincides with R_T in contrast to the absolute quantifier which profiles a specific proportion of R_T .

The standard English constructional schema [[PRONOUN] [NOUN]_{NOMINAL}] which is used four times in the idiolect data (e.g “you all” “we all”) extends to a higher order schema[[DEICTIC][NOUN]_{NOMINAL}] through schematization process which then becomes functionally equivalent to the [[PROXIMAL DEICTIC] [UNIVERSAL QUANTIFIER] [NOUN]_{NOMINAL}] constructional schema of standard Hindi forming a complex category in the idiolect resulting in the construction. Both involve the construal of a single entity as reference point, within the dominion of which scanning through heterogeneous instances forms a conceptual base for the profile of a single multiplex entity.

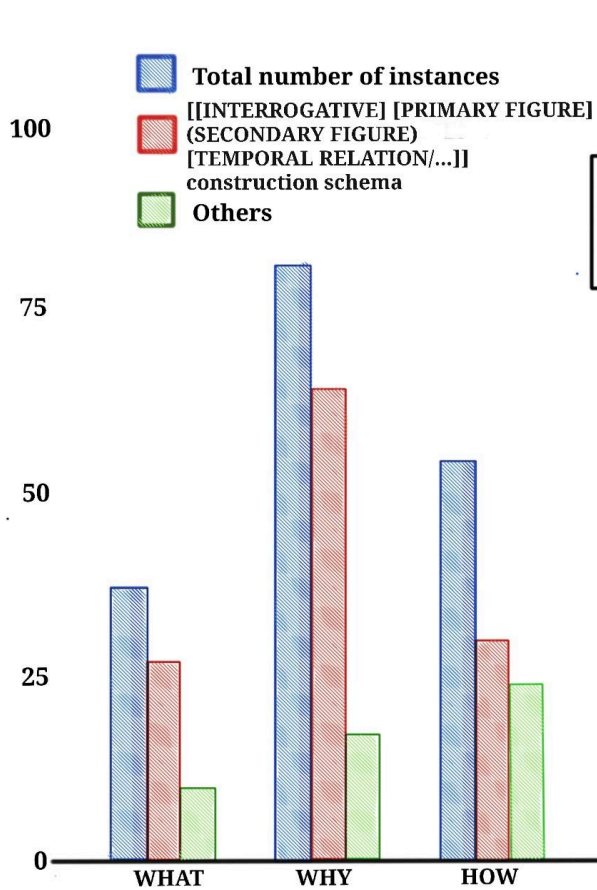
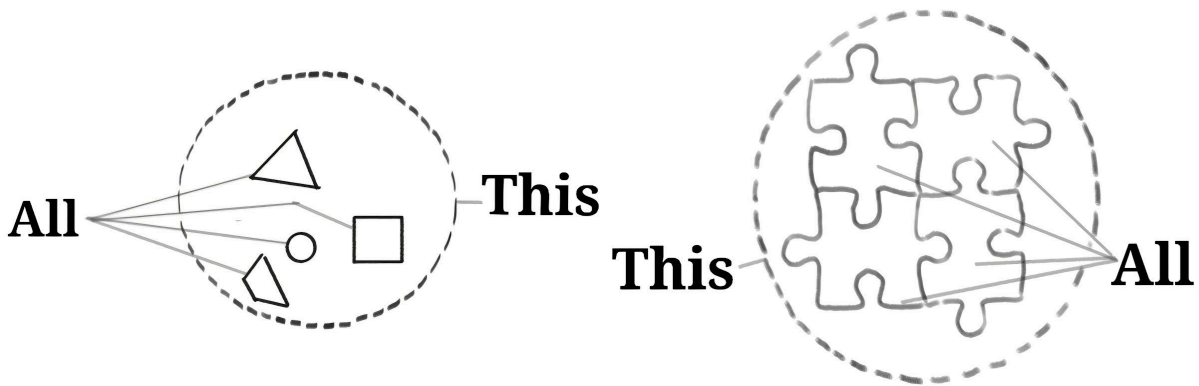
“This all has been done.”



The idiolect thus shows a contrast between [ALL PROXIMAL DEICTIC NOUN/ all these ...] and [PROXIMAL DEICTIC ALL (NOUN)/...] essentially separating the universal quantifier presupposing a region in a domain from a region in a domain presupposing the universal quantifier with respect to form meaning iconicity. The proximal deictic is acting as a superordinate grounding element which imposes summary scanning on a quantified mass by neutralizing the heterogeneity in the type space. It is a single bounded substance that is proximal to the CDS. Since the proximal deictic is used most often as an entire nominal with the noun type remaining schematic, the schematic noun serves to limit the scope of the universal quantifier to the maximal extension of the type.

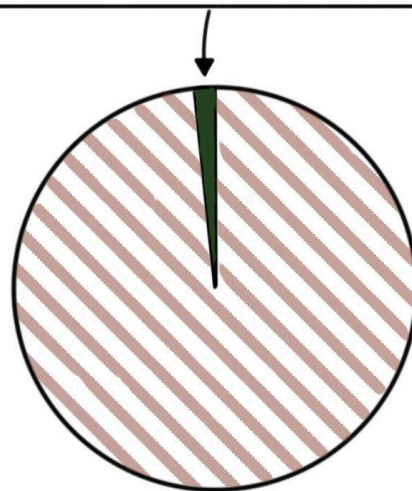
In the predicate [[ALL THIS]/[all this]], the conception of the maximal extension, does not neutralize the heterogeneity of the type space. The heterogeneity of the type space is neutralized in the concept of the proximal deictic which structures a conception at a higher schematic level. The reference point is a region, the maximal extension of the type space where the heterogeneity of the type space remains in profile with boundedness where expendability and contractibility applies only within the boundary. But the predicate, [[THIS ALL/this all]] invokes the conception of a region, where heterogeneity of instances within the region is neutralized in the proximal deictic predicate, which serves as reference point within which is located the target conception of a multiplex entity which is a bounded region where expendability and contractibility apply to the bounded region but the heterogeneity of the type space is not neutralized in the profile. These two constructional schemas form a complex category which is differentiated at a lower schematic level based on the contrast introduced in the semantic pole. There is a contrast made between 'this' and 'all' which both serve as a reference point in a

structure of which they are both part. Where ‘all’ is a reference point, ‘this’ is a target. Where ‘this’ is a reference point, ‘all’ is a target.



Total number of instances of
interrogative predicate
within clause structure
= 172

Total instances of
[INTERROGATIVE DO TENSE PARTICIPANT]
out of all instances of Predicates with
interrogative component = 3



4.2.2 The structure of the interrogative clause

The clause structure with the interrogative predicate component behaves differently. Besides the question word occupying one single attentional window, or part of a technical jargon “knowing what”, the interrogative predicate is used 172 times as a component within a clausal structure in the course of 7 lectures. There is significant constructive effort coding the standard English constructional schema [INTERROGATIVE DO TENSE PARTICIPANT]/.....].

This is evidenced by

" Why they do (pause) why do they”

Which represents a single instance out of 3 instances in which this type of structure is produced.

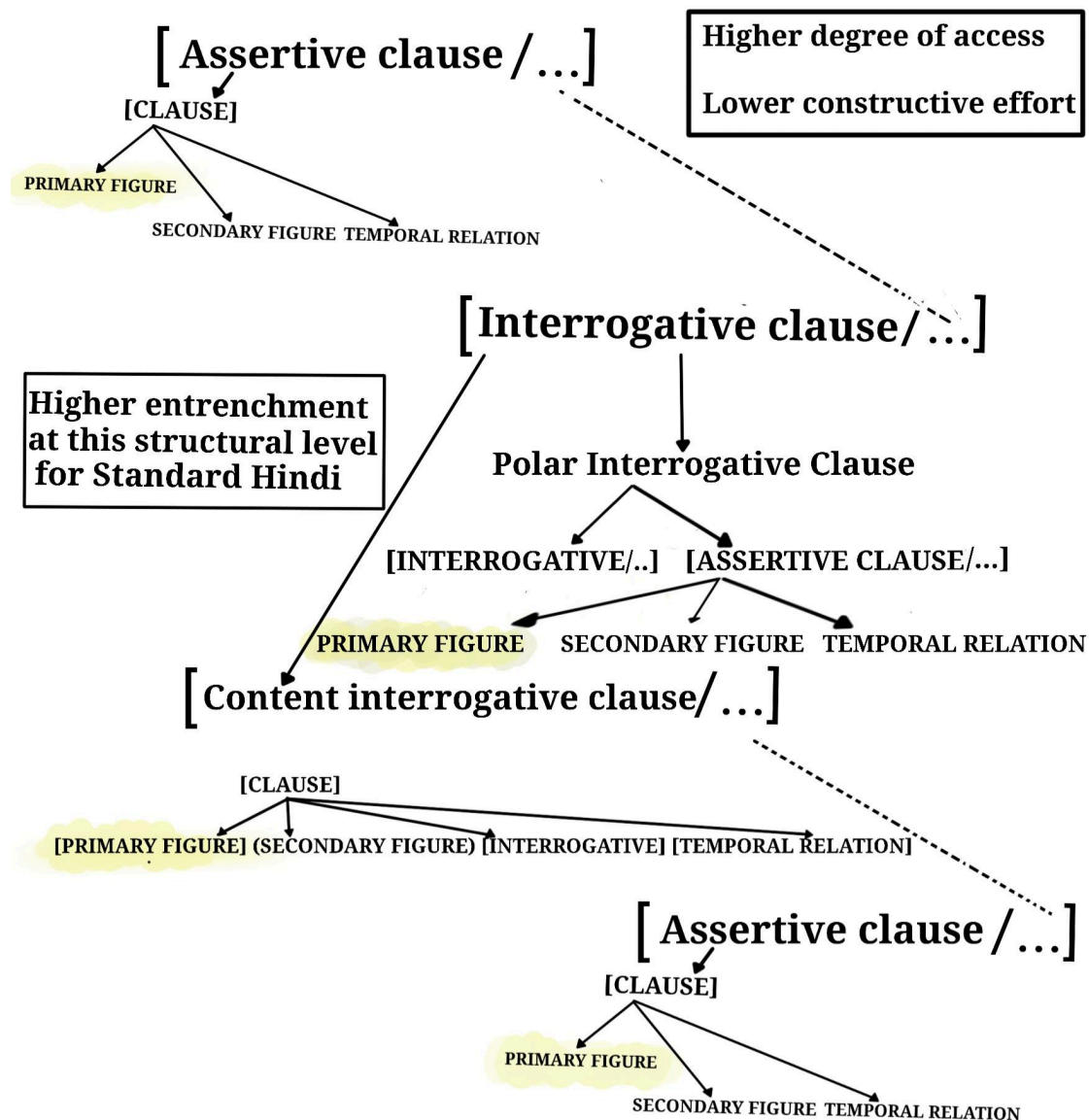
[INTERROGATIVE AGENT PARTICIPANT RELATION PARTICIPANT/...] or

[INTERROGATIVE AGENT PARTICIPANT RELATION/...] is preferred in the idiolect

In standard Hindi, the polar interrogative are questions which have assertive or negative responses as adjacent pair. This interrogative [KYA)/ kya] is placed at the beginning of the sentence as a standard for formal writing. The standard is taught repeatedly as a medium for expression. The primary figure is consistently put at the beginning of the sentence. The participants always appear before the temporal relation predicate unless the temporal relation is the brought to the beginning for emphatic purposes.

In this case, to reduce cognitive load, generalisation has led to a different type of language behaviour. The analogy has erased many functions of the standard Hindi predicates and construction schemas. Many construction schemas even if brought over into the idiolect, behave differently on a specific kind of interaction with the other standard.

The entrenchment of specific construction schemas whether because of generalization or the repeated categorization of usage events requires least constructive effort leading to this type of target structure as a resolution to the coding problem which reduces cognitive load during spontaneous pedagogical delivery for the subject.



[KYA PRIMARY FIGURE SECONDARY FIGURE TEMPORAL RELATION/ kya...] is a polar interrogative.

For a content interrogative, like kahā

Tum jā kahā rahē ho (Satyarthi et al., 2024, 2024, p. 191)

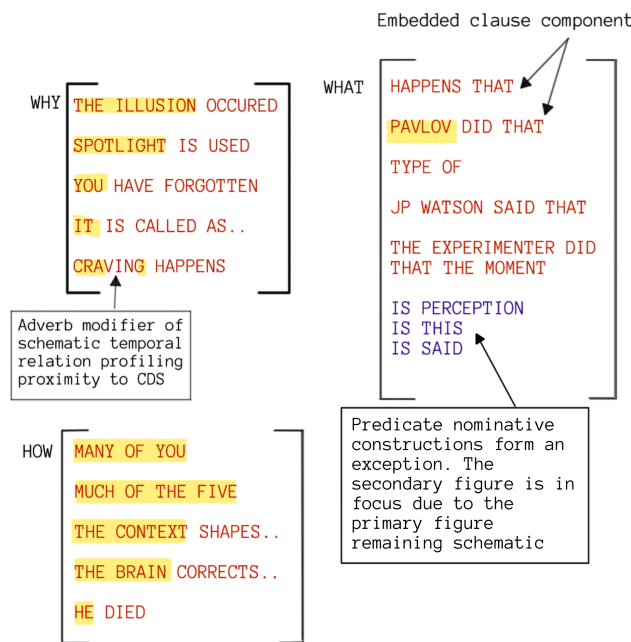
Vah gaya tha kya? (National Council of Educational Research and Training [NCERT], 2024, p. 145)

It uses the construction schema

[PRIMARY FIGURE UNGROUNDED VERB INTERROGATIVE IMPERFECTIVE
PROXIMAL TO CDS/ Tum jā kahā rahē ho]

For emphatic purposes where the question word can be placed before the ungrounded verb predicator, or between the predicator and the modifying dependents which are profile determinants at the higher compositional level.

However, the primary figure remains at the beginning of the clause. The primary participant is generally the primary figure of the clause which remains standardly in focus within the clause. This is the neutral standard. Artifactual Hindi additionally allows an emphatic function of positioning the verb after the interrogative in the sentence initial position. These characters of the artifactual unit are not entrenched enough to affect the automatization for access to contrasting instances in the type space. This neutral standard has components which serve identical functions as the other English standard. This further automatizes the configuration.



Emphatic purposes, and agent defocusing techniques aside, the idiolect maintains this construction schema with standard English artifactual predicate components. Emphatic constructions are not neutral but based on how the usage event is categorised at the instance level. English standard is constrained with regard to placement and does not allow the standard to deviate from [PRIMARY FIGURE TEMPORAL RELATION SECONDARY FIGURE/..]. This has a role in the greater frequency of primary figure being placed before the verbal component. Polar questions in English allow temporal grounding to be placed before the primary figure. However, artefactual Hindi auxiliary modifies the temporal grounding of the verb and remains in sentence final

position. With the content interrogative, the auxiliary in the English artifactual standard including Indian English, allows a schematic verbal component which is temporally grounded to appear before [PRIMARY FIGURE- VERB-TEMPORAL GROUNDING-(SECONDARY FIGURE)]. A schematic verbal component which is temporally grounded has no equivalent in the Hindi standard. The verbal modifier components

[DO IMPERFECTIVE-(AGENT)-(ADVERB)-TEMPORALLY GROUNDED TO CR/ kar rahe ho]

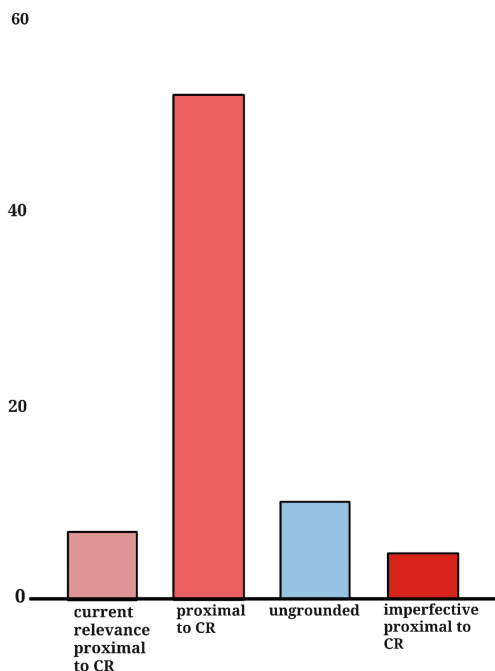
Are rarely separated in standard Hindi unless for emphatic purposes which sometimes place the agent participant between the imperfective and: adverb modifiers and the profile determinant temporal grounding. The mixed idiolect does not employ this function.

The assertive clause structure is maintained with the primary figure following the interrogative. The Hindi contrast between emphasizing the participant or the verb type or the temporal grounding is unutilized in the constructed idiolect due to entrenchment patterns.

Accessing the primary figure requires least constructional effort in the idiolect. This constructional schema is highly entrenched within the structured inventory creating a type of complex category with the artifactual English grammatical and conceptually loaded predicates.

In all the instances where the event is situated proximal to current reality, the primary figure appears before the auxiliary and the temporal relation.

Total instances of the predicate 'do' as component within clausal structure =74



With the interrogative predicate “how”, an ungrounded temporal relation appears with a passive participant in the target domain.

The primary figure mostly remains in the initial position in the constructional schema which has been automatized as evidenced by low constructional effort and easier access indicated by extremely low access for the

[[[INTERROGATIVE] [DO] [CLAUSE]]]/ how does sensation become perception]]]

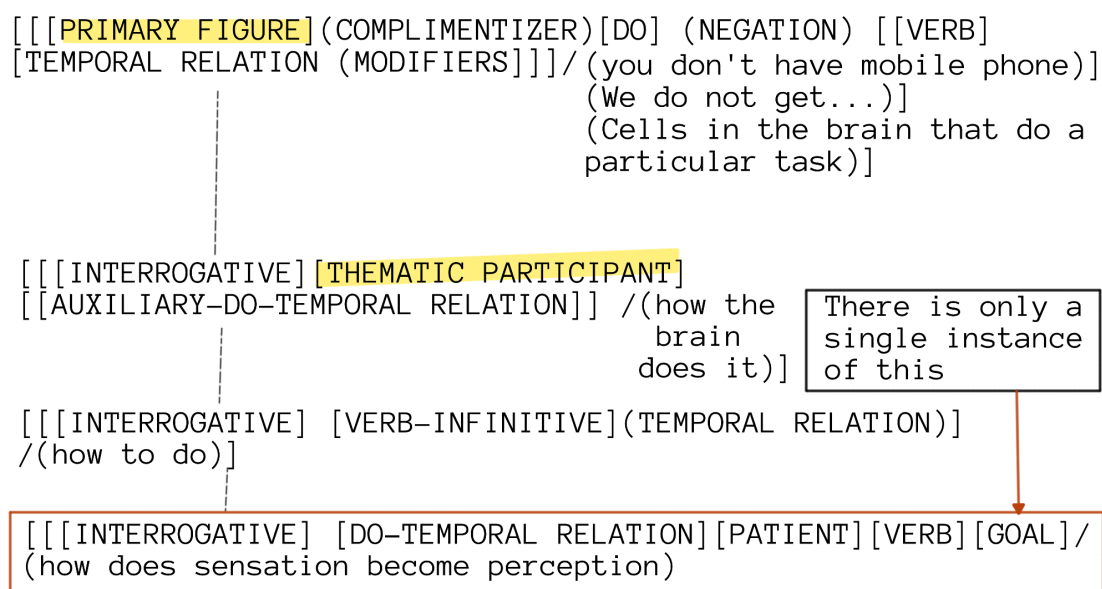
The extremely high frequency of standard structures using constructional schemas with the primary figure in the clause initial position creates extension relations between artifactual symbolic categories.

Access to the [[INTERROGATIVE] [DO-TEMPORAL RELATION] [PATIENT] [VERB] [GOAL]] construction schema is utilised when the primary figure is a thematic participant or a patient. If we observe the correction,

“Why they do.... Why do they?”

An agent is accessed with least constructive effort following the interrogative predicate indicating greater entrenchment of the artifactual constructional schema which is further forming complex categories with artifactual English construction schemas of assertive sentences and their negation.

The verb do as a predicate which can function as a schematic verb usually complemented by a dependent profile determinant which specifies the temporal grounding of the clause;



When we consider an utterance like,

“What happens that” which is syntagmatically linked to a clause, the construction schema is like the Hindi collocative construction schema

[[WHAT]- [HAPPEN-TEMPORAL GROUNDING]- [COMPLEMENTIZER]- [CLAUSE]/kya hota hai ki]] is a standard Hindi discourse marker.

"अक्सर क्या होता है कि लोग खान-पान में लापरवाही बरतते हैं, जिसका असर उनकी सेहत पर कुछ समय बाद दिखाई देता है।" (Dainik Jagran, 2023)

" Aksar Kya hotā Hai ki log khān pān mein laparvahi baratte hai, jiskā asar unki sehat per kuchh samay bad dikhāi deta hai"

"उस शाम क्या हुआ कि अचानक तेज आंधी आई और पूरा इलाका अंधेरे में डूब गया।"(Hindustan, 2022)

" Us Shām kyā huā ki Achānak Tej āndhi āi aur purā ilāka andhere Mein doob Gayā"

Both use the [[TOPIC RP INTERROGATIVE HAPPEN-TEMPORAL GROUNDING COMPLEMENTIZER [CLAUSE]]/...] construction schema which has collocation status in Hindi. This is standardised by how the press moved through the socio political cultural landscape, to adapt and create a language of mass communication which was motivated to use categories and structures used by the people in daily life.(Das,2021)

The data clearly shows predominance of artifactual Hindi constructional schemas being used with artifactual english predicates at a lower compositional level.

The entrenchment of the systematic initial occurrence of the primary figure except for predicate nominative constructions, indicates entrenchment of construction schemas which are used more often. Since the artifactual units of Hindi structured inventory, contrasts the emphatic against the neutral: the standard or neutral is far more deeply entrenched.

Functions which are within standard Hindi artifactual unit are also utilised in other cases. In the use of [[FURTHER MORE/ further more] indicating an increment stage beyond an initial point which serves as the pole (vandeloise, first book) or the reference point. It is constructed through a complex category like “this all” but impacted due to higher access to standard Hindi constructional schemas.

The predicate further is used to indicate an increment on a schematic value scale with respect to a reference point. It is an adverb. In the idiolect, the adverb is getting modified at a lower level to form a syntagmatic connection between two adverbs forming a complex category. This complex category is motivated by the construction schema [AND ALSO/ aur bhi] where bhi has a semantic pole in which [EVEN/bhi]. The structured inventory has grammatical structures to indicate increment in multiple stages. This function has been executed using English predicates.

This is evidenced by

“As rat further more went close to the lever”

This adverb predicate " further more" is used five times in the course of the seven lectures with a specific topic in the conceptual substrate which involves sequential scanning through a mental path.

“Plants se aur bhi aur more similarity further more” .

In the example, " aur bhi" ," aur more" " further more" are all used as adverbs for the imperfective which remains schematic due to lack of temporal grounding in the utterance. All these predicates indicate increment and serve the same or similar function.

The Manak Hindi kosh

“एक अव्यय जिसका प्रयोग समुच्चय, समावेश या जोर प्रकट करने के
One particle whose use same magnitude integration or emphasis appear to do for

लिए होता है”(Verma,1964)
because of happens is

Brihat Hindi kosh

“समुच्चय-बोधक अध्यय जिसका अर्थ है- साथ ही, और भी,
Same magnitude sense particle whose meaning is together EM more EM

इसके अतिरिक्त”(Prasad et al, 2016)
Than-this even more

Nalanda Vishal Shabd Sagar

" निश्चय या समूचे सूचित करने वाला एक अव्यय; अतिरिक्त;
(Navalji, 1950)
Surety or same magnitude indicate to do- causing agent one particle; even more;

समावेश।
integration

These dictionaries operate with the goal of functional equivalence.

Rajpal Hindi to English dictionary.

" भी (ind.) also, too, even, as well.” (Bahari,1989)

Oxford Hindi-English dictionary R.S McGregor,1993)

“भी [Pkt.bhia<S.api] pcle. 1.also,too, as well 2. Even” (McGregor, 1993)

This predicate, bhī, is always described as a particle where the word samucca is used to indicate similar regions occupied within a quality scale relative to any specific domain. It is also used to indicate inclusion-integration and assimilation as the use of the word samāvesh indicates. It is an inclusion association relation and more.It is used to indicate sāth hī, implying a reference point with which there is an association inclusion relation of the target. This is similar to the function of " also”,“too” and as well. There is also an emphatic function of this predicate within the artifactual structured inventory of Hindi.

The coding problem, for the increment beyond one stage of increment, is resolved by "aur bhī” "aur more” where the predicate serves to imply emphasis while "more” is used to profile increment. [(EVEN MORE)/ even more] construction schema is not chosen over [[AND] [INCORPORATION-INTEGRATION]/ [aur bhi]]. To resolve the coding problem of incorporating the scanning through different stages of increment into the construal, the

predicate “even” is not accessed but rather “further” is accessed to include the construal of a mental path that must be traversed in increments.

Discussion

In the case of “this all”, a complex category was formed because complex categories may have extension relations between instances which are very different from one another. This also shows in the construction schema used with interrogatives; which do not use Indian English construction schema. However, the English assertive construction schema having a similar function to the neutral construction schema of Hindi, neutralizes the emphatic contrasts of the Hindi artifactual unit within the idiolect. This decategorialization results in change of structure at the level of the construction schema. The complex category also shows in construction schemas of “further more” including shifts in standard configurations like the failure to access “even” to function as “bhi” because of the placement in the construction schema. The construction schemas or frames are being built to categorize very specific usage events. The condition of use, which bilingual construction grammar stresses upon, remains within the context, realistically unidentifiable due to the multiplicity of language external factors determining entrenchment patterns. These entrenchment patterns determine access.

No noticeable difference above the average constructional effort to code the target structure is visible in any of the discussed instances with regard to access time. The processing time remains very close to the average with no explicit constructional effort indicated by increase of coding time except the repair strategy. In the repair, the access time difference is revealing of entrenchment patterns.

Complex categories result from correspondence relations which may be established between predicates within the semantic pole or the phonological pole or the symbolic unit. Correspondence relations can be formed in language without the constraints of schematicity relations or subject to standard representations of association inclusion relation, identity inclusion relation or separation inclusion relation.

Thus, a type space comprising a range from the prototypical to the deviant, may have members which are prototypical or deviant in multiple categories simultaneously. This is also an impact factor for code-switching in bilingualism. While the data shows instances of insertion and alternation which are very frequent; these integrated forms show simultaneously as evidence of the fact that synchronic and diachronic pressures are simultaneously working within the language phenomenon and at varying rates with respect each type of predicate and constructional schemas. This variance is explained by entrenchment factors which are further related to sociolinguistic patterns including occupation, age, gender, social class etc.

Conclusion

This paper attempts to investigate two compromise forms created in a bilingual pedagogical discourse examined at the idiolectal level which contain alternations and code copying along with these integrated forms adopting the terminology of Johanssen, with respect to an artifactual standard. It uses a usage based perspective to do this. While the function of the composite predicates and construction schemas have a clear role in the formation of these structural forms, these show that the assumption of a matrix language is misleading in code switching phenomenon considering that it is a product of both synchronic and diachronic pressures on a dynamic human language phenomena. The structural frame itself is subject to modification in categorizing particular usage events.

The socio-political and geo - cultural impacts reside in the entrenchment patterns of the construction schemas as shown in the presence of the varieties at the idiolectal level. The construction schemas, or the artefactual frames, as it is understood by many approaches to CS phenomenon themselves fuse due to functional reasons at the level of the idiolect. Sophisticated compromise forms are created with elaborated semantic poles which remain more towards the grammatical end of the continuum than the lexical end simultaneously with insertion and alternation. Functional factors determine construction schema structure and thus entrenchment patterns also. The frequency of coding any target structure is determined by socio- linguistic patterns including occupational jargon or registers which are prevalent in age related varieties within any particular register. These may be influenced by factors used to categorise behaviours of social groups.

The structure of the constructions themselves are affected by the topic in the conceptual substrate. The coding of the target structures follows the dynamics of the discourse. This includes the various strategies of conceptual negotiation used in a pedagogical lect specifically: the topic in the conceptual substrate, has an impact on the attention management strategies which lead to intonational groups constructed to construe information accordingly. The lack of constructional effort to code the alternate frames, reveals higher frequency of processing of this type of structure whether coding or decoding considering cognitive grammar does not posit separate processes for both.

Declarations

The study involved the non-invasive audio recording and structural linguistic analysis of 7 naturalistic classroom lectures (ranging from 35–55 minutes each). Ethical clearance for this research was handled and approved in accordance with the institutional guidelines of the Institutional Ethics Committee (IEC) at the Indian Institute of Technology Kharagpur. Prior to the commencement of data collection, explicit written informed consent was obtained from the participant—an assistant professor of psychology—permitting the recording of his lectures and the subsequent academic use of the transcribed linguistic material. Students present during these extra-departmental course lectures were fully informed of the academic, non-evaluative nature of the recording, and all identifying student data or side-conversations were strictly excluded from the research corpus.

Consent for Publication

Written informed consent was obtained from the primary participant for the publication of this study, including the qualitative analysis of his specific pedagogical idiolect and the inclusion of anonymized transcript segments within the final manuscript text.

Availability of Data and Materials

The linguistic datasets generated and analyzed during the current study (specifically the fully anonymized textual transcripts of the classroom lectures) are available from the corresponding author upon reasonable request. The raw audio files are restricted and are not made publicly available to safeguard participant confidentiality and preserve the anonymity of both the subject and the residential engineering institution.

Competing Interests

The author declares no competing financial or non-financial interests that could inappropriately influence, bias, or compromise the integrity of the work submitted in this manuscript.

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